

Module 8

Grid-Following and Grid-Forming Converters (Chaps. 7-8-9 of Yazdani's Book)

Most Important Topics and Concepts

- Different methods for controlling grid-connected or grid-imposed frequency VSC systems
- Modeling and control in $\alpha\beta$ - and dq -coordinates
- Controlling active and reactive power
- Synchronization with the ac grid, design of PLL
- Dc-side voltage control
- Modeling and control of grid-forming converters

→ Phase-locked loop

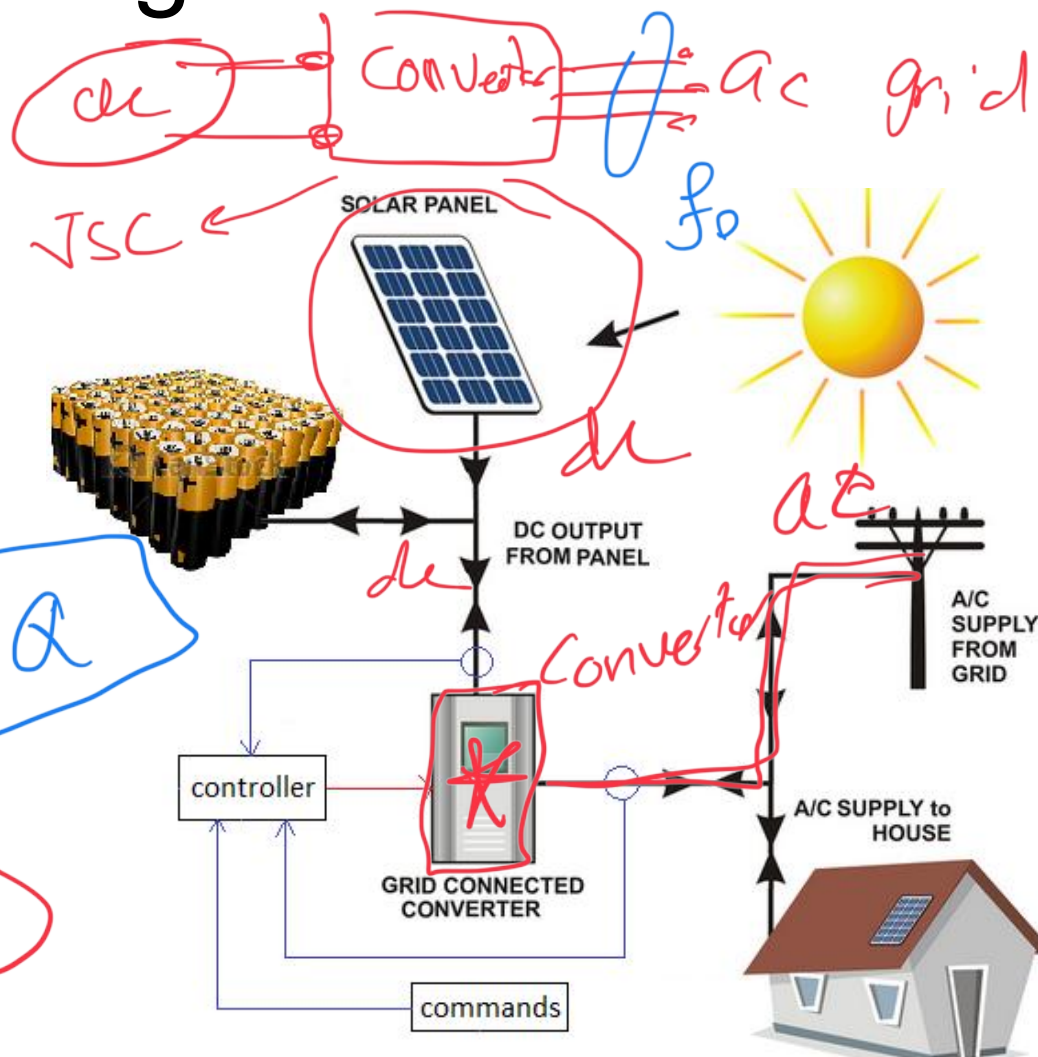
Grid-Following Converters

What we know:

- ✓ DC-side may be a DC source or another power-electronic system
- ✓ Frequency is relatively constant and imposed by the AC grid

What we want:

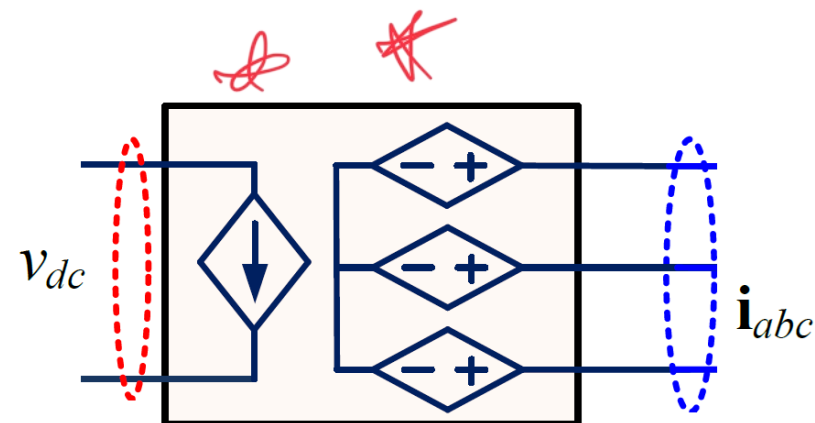
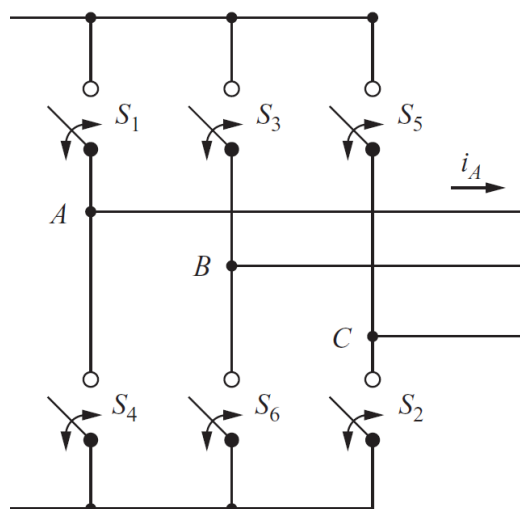
- ✓ VSC exchanges real/reactive power with AC system at PCC
- ✓ Control real/reactive power
(dc side connected to a dc voltage source)
- ✓ Control DC bus voltage
(dc side connected to a dc power source)



Three-Phase PWM Converter

❑ **Objective:** Producing 3-phase AC outputs from a DC source with PWM

❖ **3-Phase PWM Inverter:**

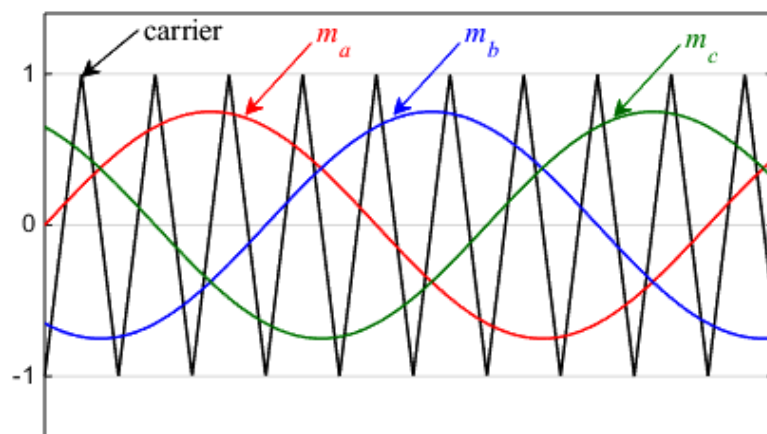


✓ Fundamental-frequency components of ac voltages:

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} m_a \\ m_b \\ m_c \end{bmatrix} \times \frac{v_{dc}}{2}$$

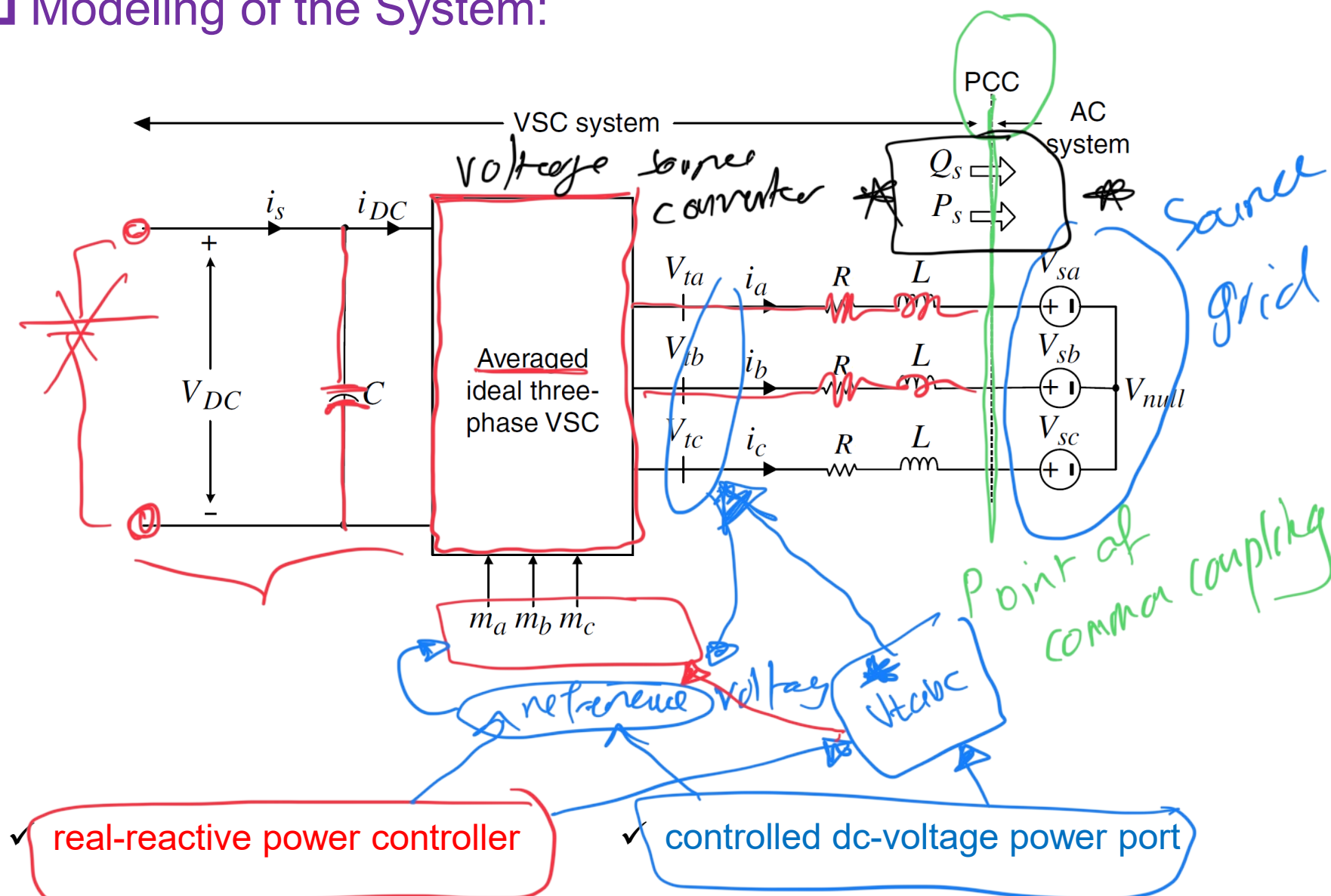
✓ Average-value of dc current:

$$i_{dc} = \frac{1}{2} \times (m_a i_a + m_b i_b + m_c i_c)$$



Grid-Following Converters

□ Modeling of the System:



Grid-Following Converters

□ Real-Reactive Power Control:

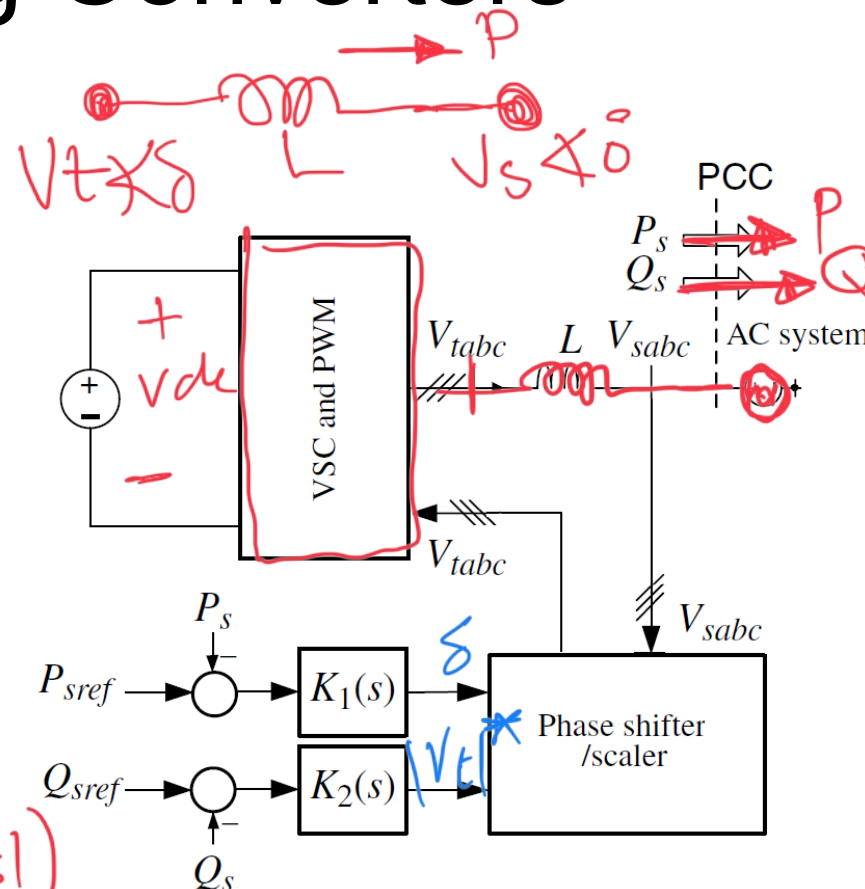
❖ Voltage-Mode Control:

$$\begin{cases} P_s = 3 \times \frac{|V_t| \times |V_s|}{X} \sin(\delta) \\ Q_s = 3 \times \frac{|V_t| \times |V_s|}{X} \left[\cos(\delta) - \frac{|V_s|}{|V_t|} \right] \end{cases}$$

Handwritten notes: $L\omega$ (above X), $V_t \cos \delta - V_s$ (above the bracket in Q_s)

small δ

$P_s \propto \delta$ $Q_s \propto (V_t - V_s)$



- ✓ P_s and Q_s may be controlled by respectively controlling the Phase Angle and Amplitude of VSC terminal voltages with respect to PCC voltages
- ✓ Simple technique with low number of control loops

✗ No control loop dedicated to currents! ✗ Currents can undergo large excursions (e.g., during faults or rapid change of commands)

❑ Real-Reactive Power Control:

❖ Current-Mode Control:

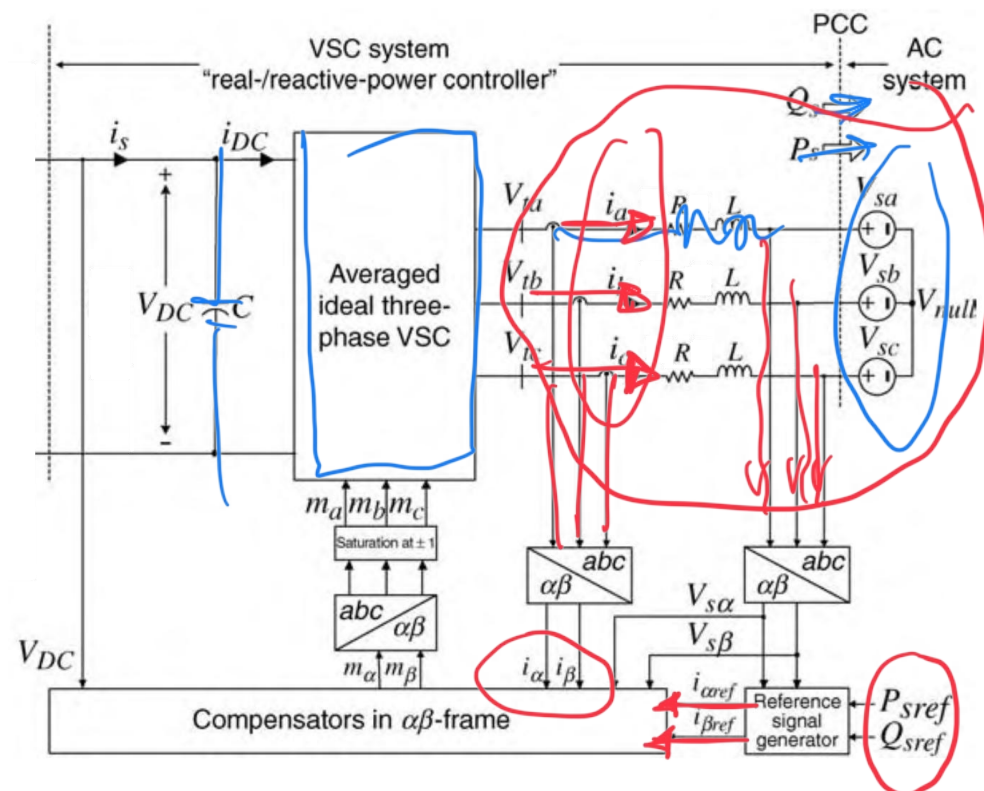
- ✓ P_s and Q_s are controlled by controlling the Phase Angle and Amplitude of VSC line currents with respect to PCC voltages
- ✓ VSC is protected against over currents because of tight current regulation
- ✓ Widely utilized for control of VSCs in power systems



- ✓ P_s and Q_s are controlled by the line current components i_α and i_β



- ✓ The reference currents i_{aref} and $i_{\beta ref}$ are limited by saturation blocks



Grid-Following Converters

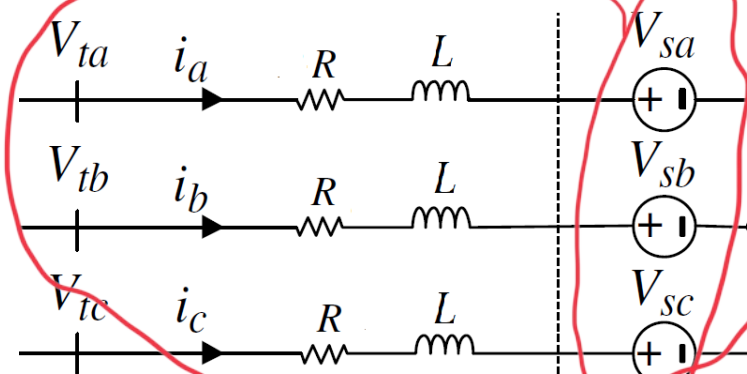
□ Dynamic Model of System:

➤ Voltages of the AC grid:

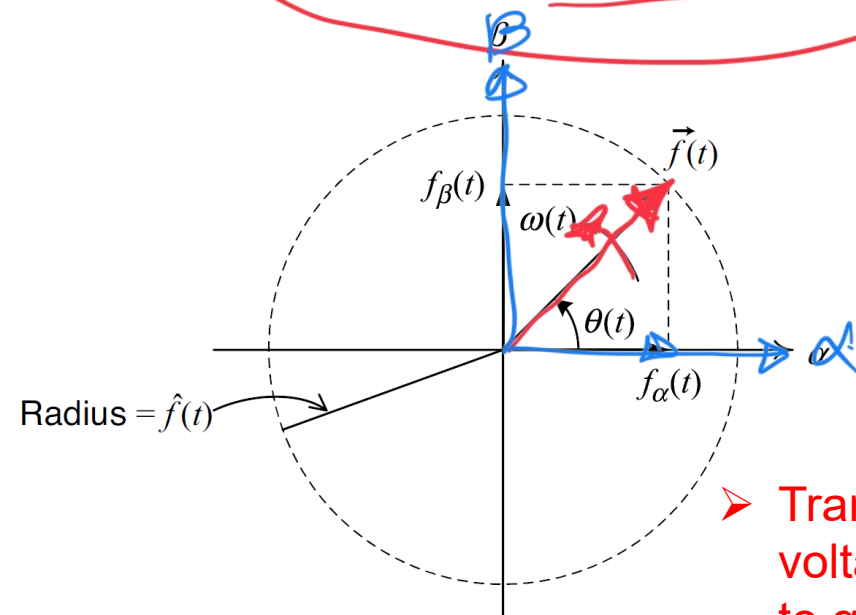
$$v_{sa}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0)$$

$$v_{sb}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 - 2\pi/3)$$

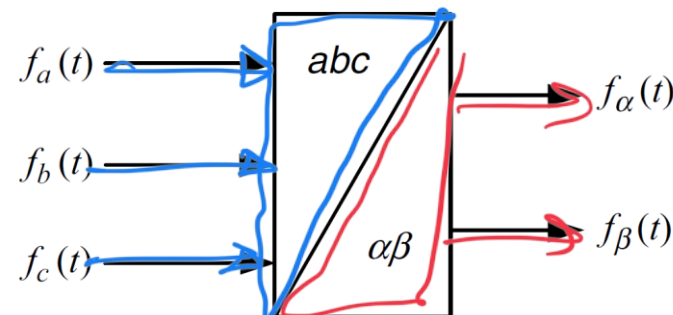
$$v_{sc}(t) = \hat{V}_s \cos(\omega_0 t + \theta_0 + 2\pi/3)$$



$$\begin{bmatrix} f_\alpha(t) \\ f_\beta(t) \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} f_a(t) \\ f_b(t) \\ f_c(t) \end{bmatrix}$$



➤ Transform the *abc* voltages/currents to $\alpha\beta$ frame

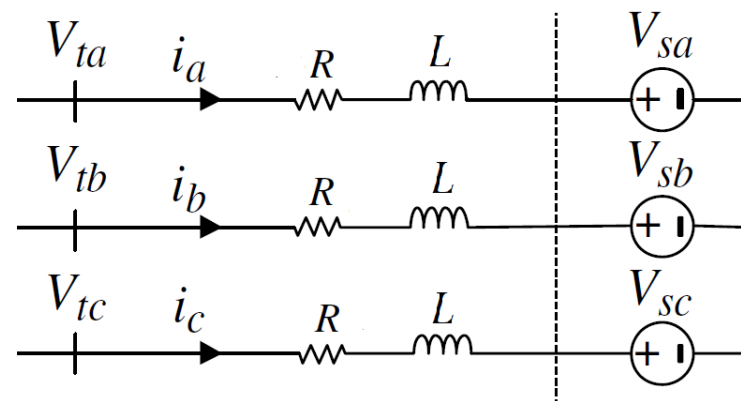


Grid-Following Converters

□ Dynamic Model of System:

➤ Active/Reactive Power Delivered to the AC grid:

$$\begin{cases} P_s = \frac{3}{2} [v_{s\alpha} i_\alpha + v_{s\beta} i_\beta] \\ Q_s = \frac{3}{2} [v_{s\beta} i_\alpha - v_{s\alpha} i_\beta] \end{cases}$$



➤ What can be controller to control P and Q ?

$$\begin{cases} i_{\alpha ref} = \frac{2}{3} \left[\frac{v_{s\alpha}}{v_{s\alpha}^2 + v_{s\beta}^2} P_{sref} + \frac{v_{s\beta}}{v_{s\alpha}^2 + v_{s\beta}^2} Q_{sref} \right] \\ i_{\beta ref} = \frac{2}{3} \left[\frac{v_{s\beta}}{v_{s\alpha}^2 + v_{s\beta}^2} P_{sref} - \frac{v_{s\alpha}}{v_{s\alpha}^2 + v_{s\beta}^2} Q_{sref} \right] \end{cases}$$

if current controller ensures

$$i_\alpha \approx i_{\alpha ref} \quad i_\beta \approx i_{\beta ref}$$



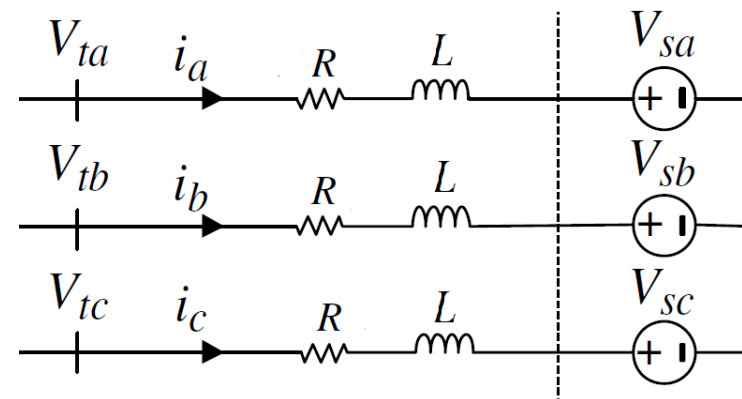
$$P_s \approx P_{sref} \quad Q_s \approx Q_{sref}$$

Grid-Following Converters

□ Dynamic Model of System:

➤ Dynamics of AC-side system:

$$\begin{cases} L \frac{di_a}{dt} = -Ri_a + v_{ta} - v_{sa} - v_{null} \\ L \frac{di_b}{dt} = -Ri_b + v_{tb} - v_{sb} - v_{null} \\ L \frac{di_c}{dt} = -Ri_c + v_{tc} - v_{sc} - v_{null} \end{cases}$$



➤ Transform the *abc* voltages/currents to $\alpha\beta$ frame

$$\begin{cases} v_{t\alpha} = \frac{V_{DC}}{2} m_{\alpha} \\ v_{t\beta} = \frac{V_{DC}}{2} m_{\beta} \end{cases}$$



$$\begin{cases} L \frac{di_{\alpha}}{dt} = -Ri_{\alpha} + v_{t\alpha} - v_{s\alpha} \\ L \frac{di_{\beta}}{dt} = -Ri_{\beta} + v_{t\beta} - v_{s\beta} \end{cases}$$

$$\begin{cases} L \frac{di_{\alpha}}{dt} = -Ri_{\alpha} + \frac{V_{DC}}{2} m_{\alpha} - v_{s\alpha} \\ L \frac{di_{\beta}}{dt} = -Ri_{\beta} + \frac{V_{DC}}{2} m_{\beta} - v_{s\beta} \end{cases}$$

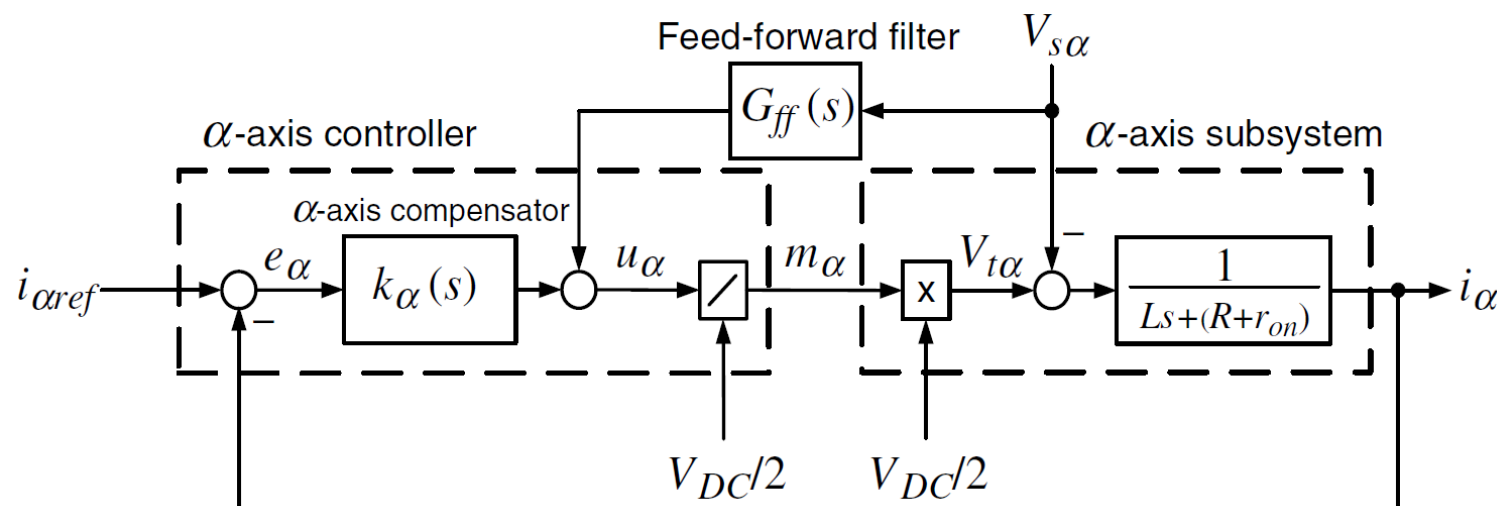
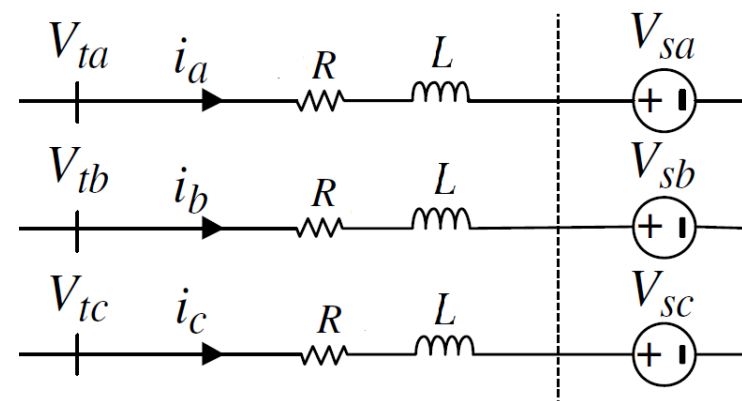


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Dynamic Model of System:

➤ Current-Controllers:

$$L \frac{di_{\alpha}}{dt} = -Ri_{\alpha} + \frac{V_{DC}}{2} m_{\alpha} - v_{s\alpha}$$

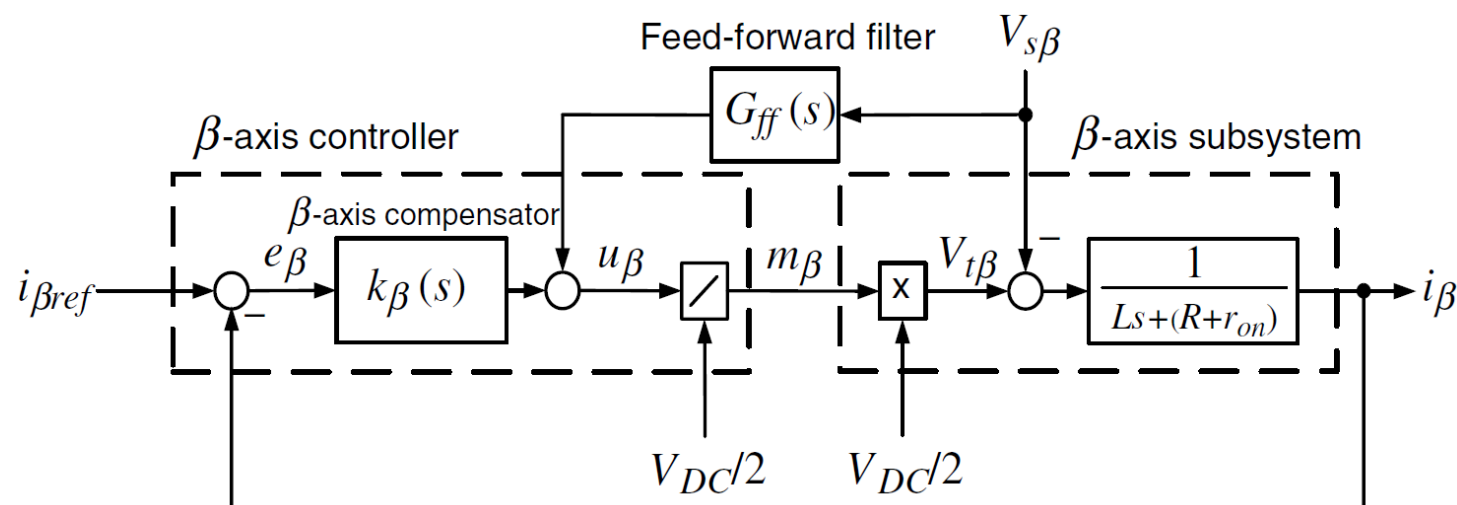
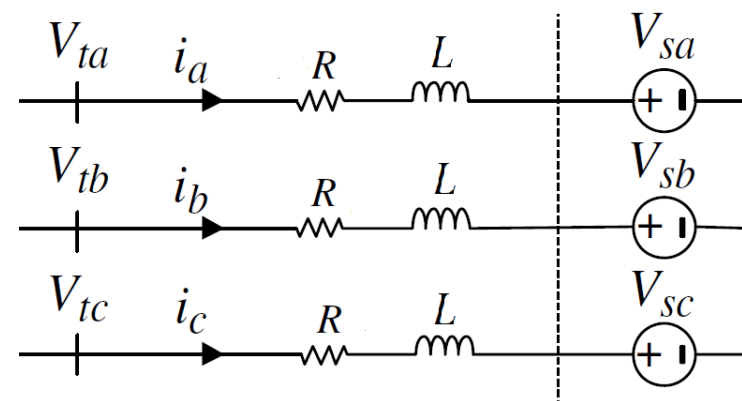


Grid-Following Converters

□ Dynamic Model of System:

➤ Current-Controllers:

$$L \frac{di_{\beta}}{dt} = -Ri_{\beta} + \frac{V_{DC}}{2} m_{\beta} - v_{s\beta}$$



Grid-Following Converters

□ Dynamic Model of System:

➤ Example 7.1:

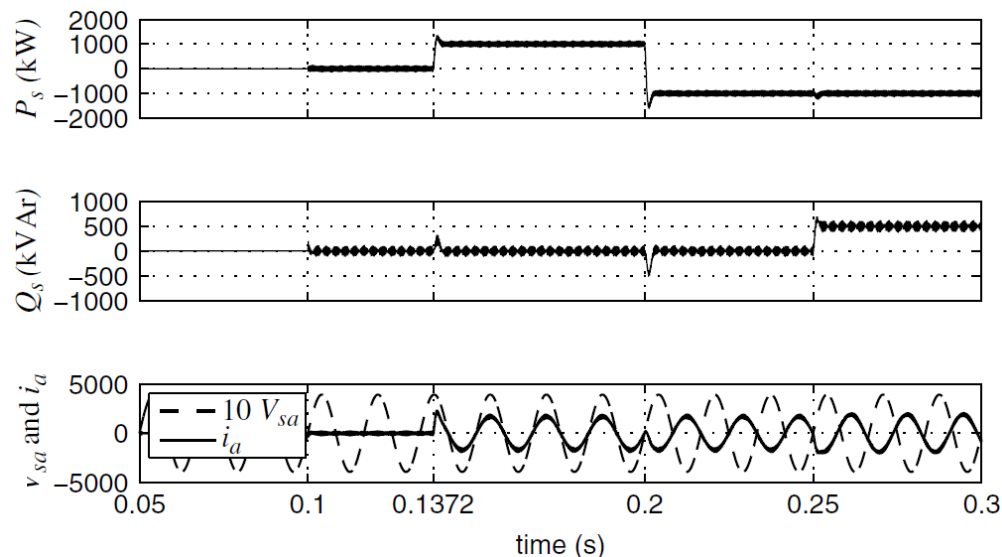
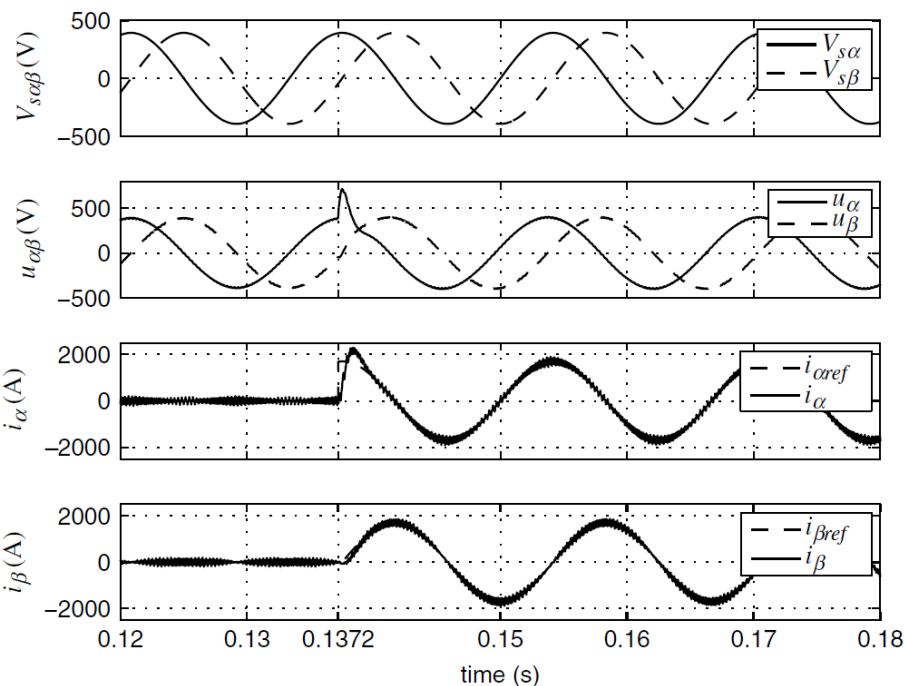
$$L = 100 \mu\text{H}$$

$$R = 0.75 \text{ m}\Omega$$

$$V_{DC} = 1450 \text{ V}$$

$$V_s^{LL} = 480 \text{ V}$$

$$\omega_0 = 377 \text{ rad/s}$$



$$\left\{ \begin{aligned} k_{\alpha}(s) &= k_{\beta}(s) = 1258 \left(\frac{s+16.34}{s^2+377^2} \right) \left(\frac{s+966}{s+5633} \right) \left(\frac{s+2}{s+0.05} \right) \\ G_{ff}(s) &= 1 / (8 \times 10^{-6} s + 1) \end{aligned} \right.$$

Grid-Following Converters

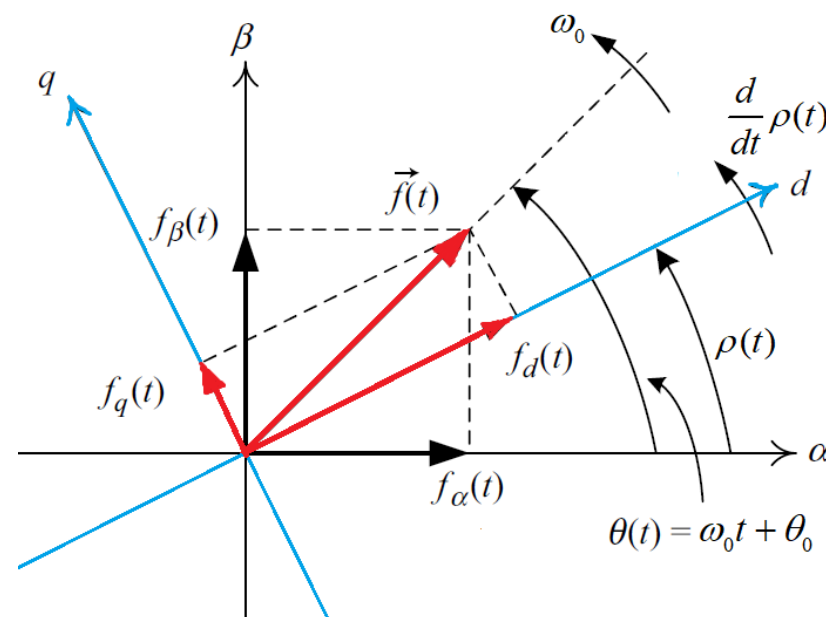
□ Space Phasor in qd -Frame:

- ✓ Consider the space phasor of any 3-phase variables:

$$\vec{f}(t) = \frac{2}{3} \left[e^{j0} f_a(t) + e^{j2\pi/3} f_b(t) + e^{-j2\pi/3} f_c(t) \right]$$



$$\vec{f}(t) = \hat{f} e^{j\theta(t)} = \hat{f} e^{j(\omega_0 t + \theta_0)}$$



- ✓ The phasor in dq -frame is expressed as:

$$f_{dq}(t) = f_d(t) + j f_q(t) = \vec{f}(t) e^{-j\rho(t)} = \hat{f} e^{j(\theta(t) - \rho(t))}$$

- ✓ Angle $\rho(t)$ can be chosen arbitrarily!

$$\frac{d}{dt} \rho(t) = \frac{d}{dt} \theta(t) = \omega_0$$



$$\begin{cases} f_d(t) = f_d \\ f_q(t) = f_q \end{cases}$$

Grid-Following Converters

□ Real-Reactive Power Control:

❖ Current-Control in dq frame:

✓ Real and reactive power in dq -frame:

$$\begin{cases} P_s(t) = \frac{3}{2} [V_{sd}(t)i_d(t) + V_{sq}(t)i_q(t)] \\ Q_s(t) = \frac{3}{2} [-V_{sd}(t)i_q(t) + V_{sq}(t)i_d(t)] \end{cases}$$

PLL in steady-state

$$V_{sq} = 0$$

$$\begin{cases} P_s(t) = \frac{3}{2} V_{sd}(t)i_d(t) \\ Q_s(t) = -\frac{3}{2} V_{sd}(t)i_q(t) \end{cases}$$

✓ P_s and Q_s are controlled by i_d and i_q respectively!

